

Finger MRI Dictation Cheat Sheet - Expanded Findings & Impressions

Copy-ready language. Replace bracketed text. Use as a workstation reference, not a substitute for surgical correlation or local report style.

Search Pattern

- Name the digit, joint, side, and zone.
- Pulley injury: report A2/A3/A4 and amount of bowstringing.
- Thumb MCP UCL: explicitly state Stener lesion present/absent.

Infection Search Pattern

- **1. Marrow/bone changes:**
 - Marrow replacement (darker than muscle/disc on T1; geographic pattern more predictive than reticular/periarticular).
 - Marrow edema (isolated finding raises concern in diabetics/high-risk).
 - Bone erosion (loss of dark T1/T2 cortical line).
- **2. Surrounding inflammatory changes:** fascia, musculature, subcutaneous fat edema/enhancement.
- **3. Susceptibility suggesting air:** in collections, soft tissues, along fascial planes, in bone.
- **4. Abnormal enhancement:**
 - Rim-enhancing collection = abscess.
 - Superficial lesion without enhancement = ulcer.
 - Enhancement along fascial planes = infectious/inflammatory fasciitis.
 - Non-enhancement of normal structures = ischemia.
 - Other patterns of enhancement may not be specific.
- **5. Differentiating osteomyelitis from neuropathic joint:**
 - Joint effusion with marrow replacement on both sides of joint.
 - Fluid collections/abscesses (periprosthetic, subperiosteal, intraosseous/Brodie's abscess).
 - Sinus tract connecting abnormal marrow signal to skin surface.
 - Disappearance of subchondral cysts on sequential imaging.
 - Progressive marrow/soft tissue changes greater than expected on sequential imaging.

Core Measurements

Structure	Normal / Threshold	Clinical Significance
Stener lesion	UCL retracted proximal and superficial to adductor aponeurosis	Surgical — ligament cannot heal in anatomic position
Pulley bowstringing	>2 mm from phalanx cortex = abnormal	Measured on axial images; >3 mm strongly suggests complete tear
Flexor zone I	FDP insertion to mid-middle phalanx	Jersey finger zone
Flexor zone II	Mid-middle phalanx to distal palmar crease	"No man's land" — FDP and FDS overlap
Flexor zone III	Distal palmar crease to distal carpal tunnel	Lumbrical origin zone
Flexor zone IV	Carpal tunnel	
Flexor zone V	Proximal to carpal tunnel	

Finger MRI Tells

- **A2 and A4** are the two most critical pulleys for finger function; A2 is at the proximal phalanx, A4 at the middle phalanx.
- **Stener lesion:** UCL retracted superficial to the adductor aponeurosis = surgical. If the ligament lies deep to the aponeurosis, conservative management may succeed.
- **FDP avulsion (Jersey finger):** ring finger most common. Look for retracted tendon stump and marrow edema at the distal phalanx base.
- **Central slip disruption** leads to boutonniere deformity (PIP flexion + DIP hyperextension) if untreated.
- **Sagittal band disruption** leads to extensor tendon subluxation at the MCP (boxer's knuckle).
- **Volar plate avulsion:** PIP most common; look for dorsal subluxation and small volar avulsion fragment.

Modified Outerbridge Chondromalacia Grading - Copy-Ready

Use this for any articular surface. Replace [surface/compartments] with the specific location: patellar median ridge, medial femoral condyle weightbearing surface, anterosuperior acetabulum, radiocapitellar joint, tibiotalar dome, first MTP joint, etc.

Grade	MRI shorthand
0	Normal cartilage.
1	Signal heterogeneity/softening or swelling; surface intact.
2	Superficial fissuring/fraying/partial-thickness defect <50% cartilage depth.
3	Deep fissuring/partial-thickness defect >50% cartilage depth, not exposing bone.
4	Full-thickness cartilage loss with exposed subchondral bone, often edema/cysts.

Grade 0 - normal cartilage

F: Articular cartilage of the [surface/compartments] is preserved in thickness and signal without focal fissuring, delamination, or subchondral marrow abnormality.

I: No chondral defect of the [surface/compartments].

Grade 1 - chondral signal change / softening

F: Mild focal cartilage signal heterogeneity/softening of the [surface/compartments] with preserved cartilage thickness and intact articular surface. No subchondral marrow edema.

I: Grade 1 chondromalacia of the [surface/compartments].

Grade 2 - superficial partial-thickness chondral loss

F: Superficial chondral fissuring/fraying of the [surface/compartments] involving less than 50% of cartilage thickness. No full-thickness defect or subchondral marrow edema.

I: Grade 2 chondromalacia of the [surface/compartments].

Grade 3 - deep partial-thickness chondral loss

F: Deep partial-thickness chondral fissuring/defect of the [surface/compartments] involving greater than 50% of cartilage thickness without exposed subchondral bone. [Mild adjacent subchondral marrow edema.]

I: Grade 3 chondromalacia of the [surface/compartments].

Grade 4 - full-thickness chondral loss

F: Full-thickness cartilage loss/fissuring of the [surface/compartments] measuring [X] x [Y] mm with exposed subchondral bone and [subchondral marrow edema/cystic change]. [Small unstable chondral flap/loose body if present.]

I: Grade 4 chondromalacia/full-thickness chondral defect of the [surface/compartments].

Normal / Negative Templates

Normal finger

F: The flexor and extensor tendons of the [digit] are intact without tendinosis, tear, or tenosynovitis. The A2 and A4 pulleys are intact without bowstringing. The collateral ligaments and volar plates of the MCP, PIP, and DIP joints are intact. The articular cartilage is preserved. No marrow edema, fracture, or soft tissue mass. No joint effusion or synovitis.

I: Normal MRI of the [digit].

Useful Caveats

- **Collateral ligament proper vs accessory:** the proper collateral ligament is the primary restraint to lateral stress and is taut in flexion; the accessory collateral ligament inserts on the volar plate and is taut in extension. Report which component is injured.
- **Flexor tendon zones:** surgical planning and prognosis depend on zone of injury. Zone II ("no man's land") tears carry the highest risk of adhesion and poor outcome.
- **Bipartite sesamoid vs fracture** at the thumb MCP: bipartite has smooth corticated margins; fracture has irregular edges with marrow edema.

Pulley / Tendon / Ligament

A2 pulley tear

F: Disruption/attenuation of the A2 pulley at the proximal phalanx with volar bowstringing of the flexor tendons measuring [X] mm from the proximal phalanx cortex, and surrounding soft tissue edema. The flexor tendons are [intact/show secondary tendinosis].

I: A2 pulley tear with flexor tendon bowstringing.

A4 pulley tear

F: Disruption of the A4 pulley at the middle phalanx with focal flexor tendon bowstringing measuring [X] mm from the middle phalanx cortex and peripulley edema.

I: A4 pulley tear.

Multiple pulley tear (A2 + A3)

F: Combined disruption of the A2 and A3 pulleys with diffuse volar bowstringing of the flexor tendons measuring [X] mm from the phalanx cortex extending from the proximal phalanx to the PIP joint level. Surrounding soft tissue edema and tenosynovial fluid.

I: Combined A2 and A3 pulley tears with flexor tendon bowstringing. Surgical consultation recommended given multi-pulley involvement.

FDP avulsion (Jersey finger)

F: Avulsion of the flexor digitorum profundus tendon from its distal phalanx insertion in the [ring/other] finger with proximal retraction of the tendon stump to the level of the [DIP joint/PIP joint/proximal phalanx/palm]. Marrow edema at the distal phalanx base. [Small volar avulsion fragment at the distal phalanx.]

I: FDP avulsion (Jersey finger) of the [digit] with tendon retraction to the level of the [location]. [Leddy and Packer type I/II/III.]

FDS laceration

F: [Partial/complete] discontinuity of the flexor digitorum superficialis tendon at the level of the [zone] with [X] mm gap/retraction and surrounding tenosynovial fluid. The FDP tendon is [intact/also injured].

I: [Partial/complete] FDS tendon tear at zone [zone] of the [digit].

Flexor tendon tear

F: [Partial/full-thickness] tear of the [FDP/FDS] tendon at zone [I-V] with [X] mm tendon gap/retraction and tenosynovial fluid. [Intratendinous signal change consistent with tendinosis at the tear margin.]

I: [Partial/full-thickness] [FDP/FDS] tendon tear at zone [zone].

Central slip tear

F: Discontinuity/irregularity of the central slip insertion at the dorsal base of the middle phalanx with surrounding edema and [small avulsion fragment]. [Lateral band subluxation to the volar aspect of the PIP joint axis, compatible with early boutonniere mechanism.]

I: Central slip extensor tendon injury. [Clinical correlation for boutonniere deformity.]

Extensor sagittal band tear

F: Disruption of the [radial/ulnar] sagittal band at the [MCP] joint with [ulnar/radial] subluxation of the extensor tendon on axial images. Surrounding soft tissue edema and small joint effusion.

I: [Radial/ulnar] sagittal band tear at the [digit] MCP joint (boxer's knuckle) with extensor tendon subluxation.

Volar plate injury

F: [Partial/full-thickness] tear of the volar plate at the [PIP/MCP] joint with capsular edema and [small volar avulsion fragment]. [Dorsal subluxation of the phalanx.]

I: [Partial/full-thickness] volar plate injury of the [joint].

Thumb MCP UCL tear (Stener lesion present)

F: Complete tear of the thumb MCP ulnar collateral ligament at the distal attachment with proximal retraction of the ligament stump superficial to the adductor aponeurosis, consistent with a Stener lesion. Adjacent soft tissue edema and small joint effusion.

I: Complete thumb MCP UCL tear with Stener lesion. Surgical repair indicated.

Thumb MCP UCL tear (non-Stener)

F: [Partial/complete] tear of the thumb MCP ulnar collateral ligament at the [proximal/distal] attachment. The ligament remains deep to the adductor aponeurosis without Stener lesion. Surrounding edema and small joint effusion.

I: [Partial/complete] thumb MCP UCL tear without Stener lesion. [May be amenable to conservative management pending clinical correlation.]

Thumb MCP RCL tear

F: [Partial/complete] tear of the thumb MCP radial collateral ligament at the [proximal/distal] attachment with periligamentous edema and small joint effusion. [No significant lateral subluxation.]

I: [Partial/complete] thumb MCP radial collateral ligament tear.

Collateral ligament injury (PIP)

F: [Partial/complete] tear of the [radial/ulnar] collateral ligament of the [digit] PIP joint at the [proximal/distal] attachment with periligamentous edema and small joint effusion. [Mild lateral subluxation on coronal images.]

I: [Partial/complete] [radial/ulnar] collateral ligament tear of the [digit] PIP joint.

Soft Tissue Masses

Giant cell tumor of tendon sheath

F: Well-circumscribed, lobulated soft tissue mass along the [flexor/extensor] tendon sheath of the [digit] measuring [X] x [Y] x [Z] mm, demonstrating low-to-intermediate T1 and low T2 signal with susceptibility artifact from hemosiderin deposition. [Mild enhancement.] No osseous invasion or aggressive features.

I: Soft tissue mass along the [digit] tendon sheath, characteristic of giant cell tumor of tendon sheath (GCTTS/PVNS). Excision may be considered if symptomatic.

Mucous cyst

F: Small, well-defined cystic lesion at the dorsal aspect of the [digit] DIP joint measuring [X] mm, demonstrating T1 low signal and T2 high signal, contiguous with the DIP joint and adjacent to a marginal osteophyte. [Mild overlying skin thinning.]

I: Mucous cyst (digital myxoid cyst) of the [digit] DIP joint associated with underlying osteoarthritis.

Enchondroma

F: Well-defined intramedullary lesion in the [proximal/middle] phalanx of the [digit] measuring [X] mm, demonstrating lobulated T1 low/intermediate signal and T2 high signal with characteristic chondroid matrix. No cortical destruction, periosteal reaction, or soft tissue extension. [Faint internal mineralization.]

I: Enchondroma of the [digit] [proximal/middle] phalanx. Benign appearance. [If pain is present, consider pathologic fracture risk.]

Source Notes

Built as a practical dictation aid from standard MSK MRI reporting patterns, review-level radiology references, and the existing shoulder cheat-sheet style. Use surgical correlation and local report conventions when needed.